From Linear Scores to a Single Hidden Layer

A Mathematical Study of Simple Learning Systems

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The Central Question & Experimental Fairness

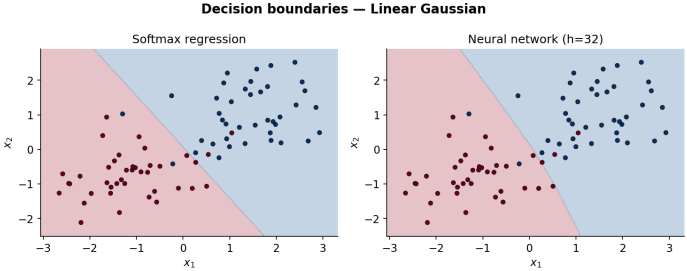
The Central Question: *When does a one-hidden-layer nonlinear classifier genuinely improve on a linear decision rule, and when is additional model complexity unnecessary?*

Ensuring a Fair Comparison: To scientifically compare Softmax Regression with our Neural Network ($h = 32$), we maintained strict experimental parity:

- **Data Parity:** Exact same train/validation/test splits across all models.
- **Preprocessing:** No hidden advantages; digits data used strictly as provided.
- **Model Selection:** Configurations and epochs were chosen using *validation cross-entropy only*. The test set was touched exactly once.
- **From Scratch:** Both models, forward/backward passes, and optimizers were built entirely in NumPy (`lite_torch`).

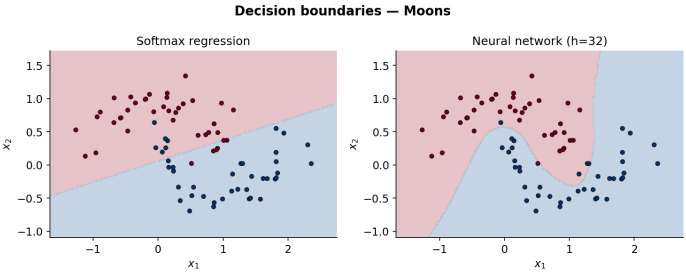
Where Linear Structure is Sufficient: Gaussian Task

- **Softmax Regression (Left):** Perfectly isolates the two clusters with a clean, affine hyperplane.
- **Neural Network (Right):** Achieves separation, but adds unnecessary nonlinear wiggles.
- **Conclusion:** When underlying data is linearly separable, added capacity begins fitting local noise. Extra complexity is unjustified.



Where Nonlinearity Helps: Moons Task

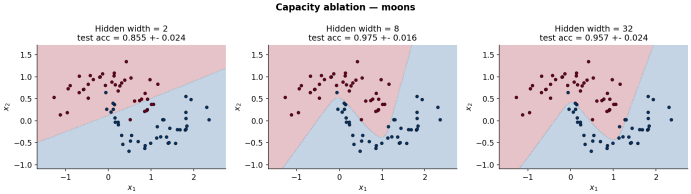
- Softmax fails completely on interlocking crescent geometry.
- **Representational Change:** The hidden layer's tanh activations geometrically warp the input space.
- This twist transforms the tangled crescents into a linearly separable representation for the final output layer.



Capacity Ablation: When Complexity Fails

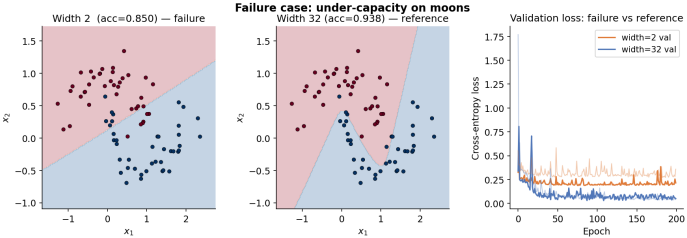
We ablated hidden width (h) on the moons task:

- $h = 2$**:** Geometric under-capacity. Fails to wrap the curves.
- $h = 8$**:** The sweet spot. Captures the true signal cleanly.
- $h = 32$**:** Over-capacity. Boundary becomes jagged, memorizing training noise and reducing generalization.



Failure Case Analysis ($h = 2$ on Moons)

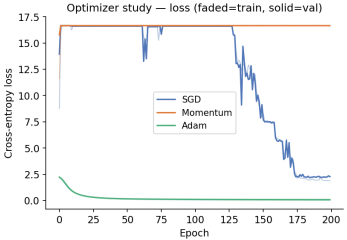
- **The Failure:** The $h = 2$ model plateaus at a high validation loss.
- **The Lesson:** This is a *structural* failure, not an *optimization* failure.
- Mathematically, two tanh neurons define only two hyperplanes. Two straight lines are geometrically insufficient to enclose a crescent shape.

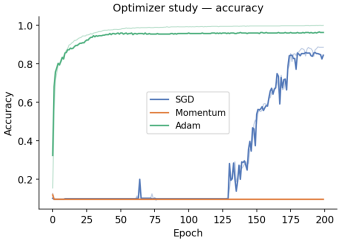


Optimization Dynamics on the Digits Benchmark

While linear models are convex, neural networks have complex loss landscapes.

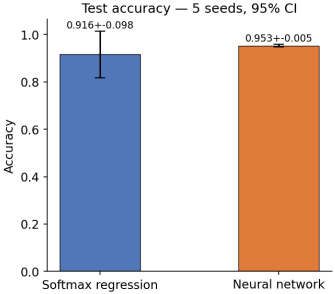
- **SGD:** Trapped in a flat saddle point for 125 epochs before escaping.
- **Momentum:** Fails to converge; high learning rate causes violent overshooting.
- **Adam:** Adaptive learning rates effortlessly navigate the landscape, converging almost instantly.

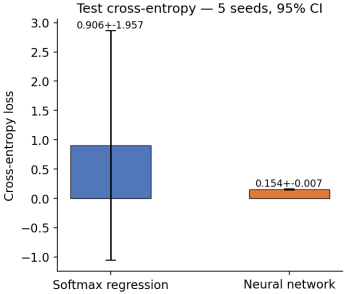




What Repeated-Seed Evaluation Adds

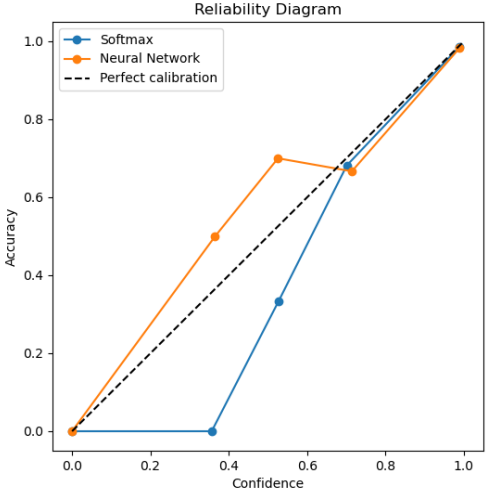
- We ran 5 random seeds for our final configurations to isolate architecture from "lucky" initialization.
- **NN:** 0.953 ± 0.005
- **Softmax:** 0.916 ± 0.098
- **Conclusion:** The tight CI for the neural network proves its superiority is statistically significant and highly robust.





Advanced Track: Prediction Confidence & Reliability

- **The Trade-off:** The neural network gains accuracy but loses calibration.
- To minimize cross-entropy, the massive capacity of the hidden layer pushes logits to extreme values.
- The NN becomes severely overconfident, acting more like a ranking score than true probability.



Discussion & Conclusions

1. **When is linear sufficient?** When the data is linearly separable (Gaussian) or when perfectly calibrated probability is prioritized over raw accuracy.
2. **Representational Change:** The hidden layer bends and warps the input space so that entangled data becomes linearly separable at the output layer.
3. **Unjustified Complexity:** Over-capacity (e.g., $h = 32$ on Moons) causes the model to fit statistical noise, creating jagged boundaries and hurting generalization.
4. **The Failure Case:** Optimization algorithms cannot overcome a fundamental lack of geometric capacity ($h = 2$).
5. **Repeated Seeds:** Proves that our architectural gains are a mathematically robust reality, not an artifact of random chance.

Limitations of the Study

To maintain scientific honesty, we acknowledge the following boundaries of our study:

- **Shallow Architecture:** We evaluated a single hidden layer. Deep networks build hierarchical representations and may exhibit entirely different optimization plateaus.
- **Activation Constraints:** We strictly used tanh. Modern architectures heavily favor ReLU, which mitigates the vanishing gradient problem that likely trapped our SGD optimizer on the digits task.
- **Sample Size in Reliability:** Our calibration analysis used a small test set, leading to high variance in the middle-confidence bins. These calibration conclusions are preliminary.

Thank You!

Q&A Part